REVERSE ENGINEERING

Activity Report

TWS Earbuds & Charging Case
Power IC under study: IP5416P



What this report covers

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Part I

Studying the Product

- External features & construct
- Subsystems & modes of operation
- Purpose & user needs

II

Part II

Origin to Latest Inventions

- Technology across versions
- Core smart technologies today
- Why the evolution matters

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Part III

Internal Components & Flow

- Teardown photographs
- Block diagram & subsystems
- Working flow, BOM & specs

Part I
Studying the Product

External features, subsystems, modes of operation, purpose



External features & physical construct



Pill-shaped white case

Hinged lid, rounded edges, pocket-sized

USB Type-C charging port

Single wired power input to the case

Stem-style earbuds

Silicone ear tips seal the ear canal

Magnetic lid closure

Magnet seat in lid and base shell

Subsystems used

Power input

USB Type-C, 5 V

Power management

IP5416P charge + boost SoC

Energy storage

Li-polymer cells in case and buds

Wireless audio

Bluetooth SoC in each earbud

Transducers

Dynamic driver and microphones

Mechanics

Shells, hinge, latch, magnet

Modes of operation

1



Case charging

Type-C 5 V fills the case battery.

2



Standby

Lid closed, IC sleeps to save energy.

3



Earbud charging

Lid open, boost feeds the buds.

4



Playback & calls

Buds out, Bluetooth audio runs.

Purpose & user needs addressed

PURPOSE

Play audio wirelessly and recharge on the go from a pocket-sized case.



Wire-free freedom

No cable between ears or phone

All-day power

Case recharges buds several times

Portability

Compact case fits any pocket

Safe recharge

Protection against over-charge, short

Part II
Product Origin to Latest Inventions

How TWS charging cases evolved and why it matters



Technology across versions

Early stage

IMPROVEMENT

First wire-free buds, simple power-bank case

DRAWBACK

Bulky, many parts, little battery info

First generation

IMPROVEMENT

Charger and boost in one IC, LED level

DRAWBACK

No smart control, basic protection

Second generation

IMPROVEMENT

Switching charger, MCU, Type-C, lid detect

DRAWBACK

Higher standby drain, many support parts

Latest generation

IMPROVEMENT

One SoC: charge, boost, protection (IP5416P)

DRAWBACK

Locked to one vendor chip

Current status: core smart technologies

Type-C input	Reversible, standard 5 V case charging
IP5416P SoC	Charge management, boost and protection in one chip
Contact charging	Pins power both earbuds when inserted
Lid detection	Magnet wakes the IC on open, sleeps on close
Bluetooth earbuds	Own SoC, microphones and driver in each bud

Why this evolution matters

Fewer parts

One SoC replaces separate charger, boost and control ICs, so the board shrinks.

Higher efficiency

Synchronous boost wastes less energy, so more earbud recharges per case charge.

Safer

Built-in over-charge, over-current, short-circuit and temperature protection.

Longer standby

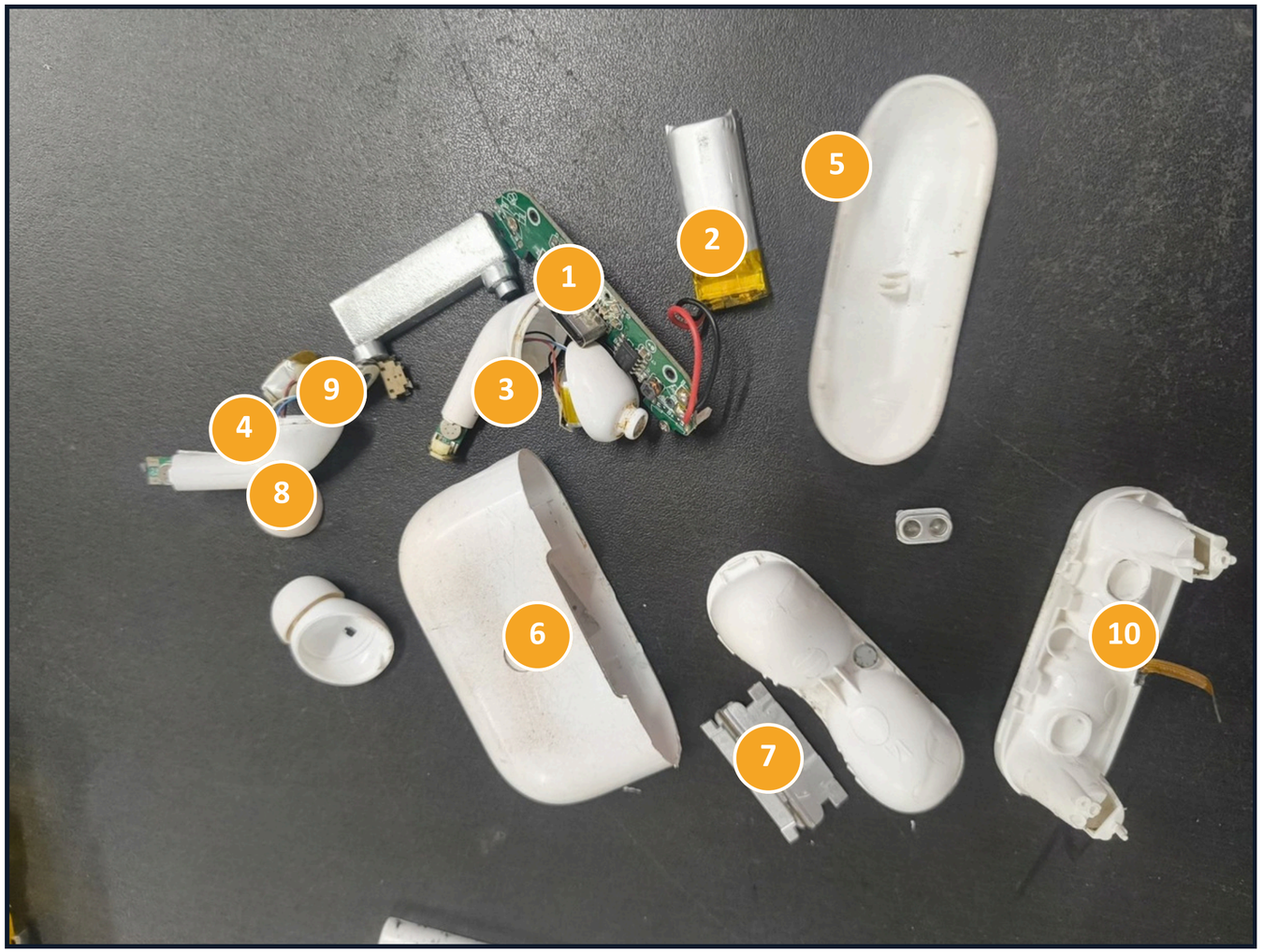
Sleep mode keeps the case battery from draining on the shelf.

Part III
Internal Components & Working Flow

Teardown, block diagram, subsystems, bill of material



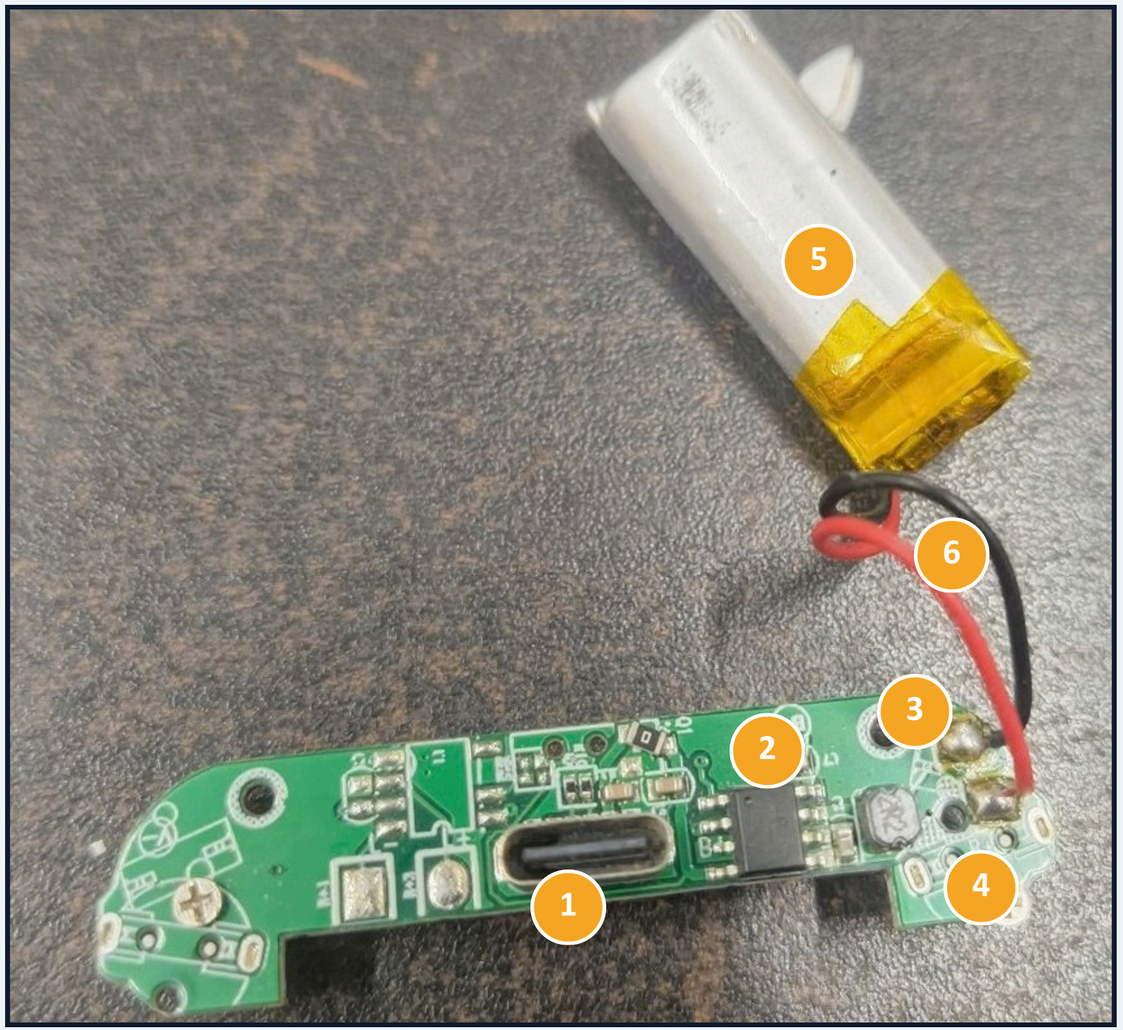
Inside the product: full teardown



1. Case PCB + Type-C
2. Case Li-polymer cell
3. Earbud 1 housing
4. Earbud 2 housing
5. Case lid shell
6. Case base shell
7. Hinge / latch metal part
8. Silicone ear tip
9. Dynamic driver
10. Shell with flex cable

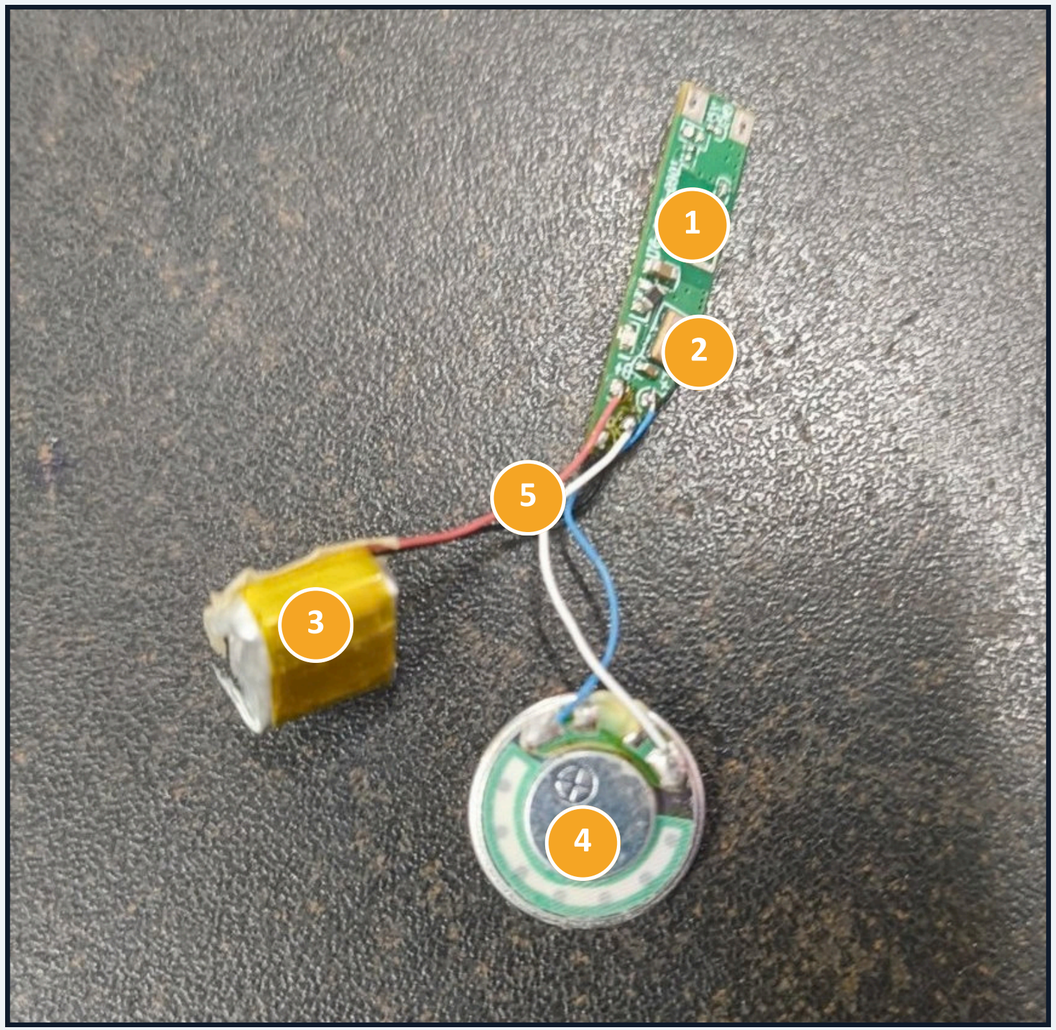
Marker positions are approximate.

Charging-case electronics



1. **USB Type-C port**
   5 V wired input to the case
2. **IP5416P (8-pin IC)**
   Charge management and boost; confirm marking
3. **Boost inductor**
   Energy storage for the boost converter
4. **Battery solder pads**
   Red and black leads from the cell
5. **Li-polymer cell**
   Case energy store, 3.7 V nominal
6. **Battery leads**
   Power path from cell to board

Earbud electronics



1. **Earbud PCB**
   Bluetooth electronics, antenna trace and charge pads
2. **Small components**
   Passives and module on the board
3. **Li-polymer cell**
   Powers the earbud; yellow tape wrap
4. **Dynamic driver**
   Round speaker unit converts signal to sound
5. **Lead wires**
   Cell, driver and PCB interconnect

External parts explained

Lid shell
Top cover; guards the earbuds and holds a magnet

Base shell
Holds the PCB, cell and the two earbud wells

Hinge + latch
Metal parts that let the lid open and snap shut

Magnet seat
Round pocket for the lid-closing magnet

Type-C opening
Cut-out giving access to the charging port

Earbud housing
Stem-style plastic shell protecting the electronics

Silicone ear tip
Seals the ear canal for comfort and isolation

Inner shell + flex
Rear cover with a thin flex cable (role [TBC])

Acoustic mesh part
Small dual-port metal piece, likely a vent [TBC]

Internal parts: charging case

Case PCB

Green board carrying the Type-C port, IP5416P and passives

Type-C receptacle

5 V power input; CC resistors identify the charger

IP5416P SoC

Charges the cell, boosts to 5 V for the buds, adds protection

Boost inductor

Stores energy in the boost step and smooths the current

Resistors + capacitors

Filter noise, set currents and steady the voltage rails

Li-polymer cell + leads

3.7 V cell in yellow tape; red and black leads soldered to pads

Internal parts: earbud

Earbud PCB

Bluetooth electronics, antenna trace and charge pads on a slim board

Earbud cell

Tiny 3.7 V Li-polymer cell that powers the bud

Dynamic driver

Coil, magnet and diaphragm turn the signal into sound

MEMS microphone

Tiny sensor on the stem picks up your voice for calls

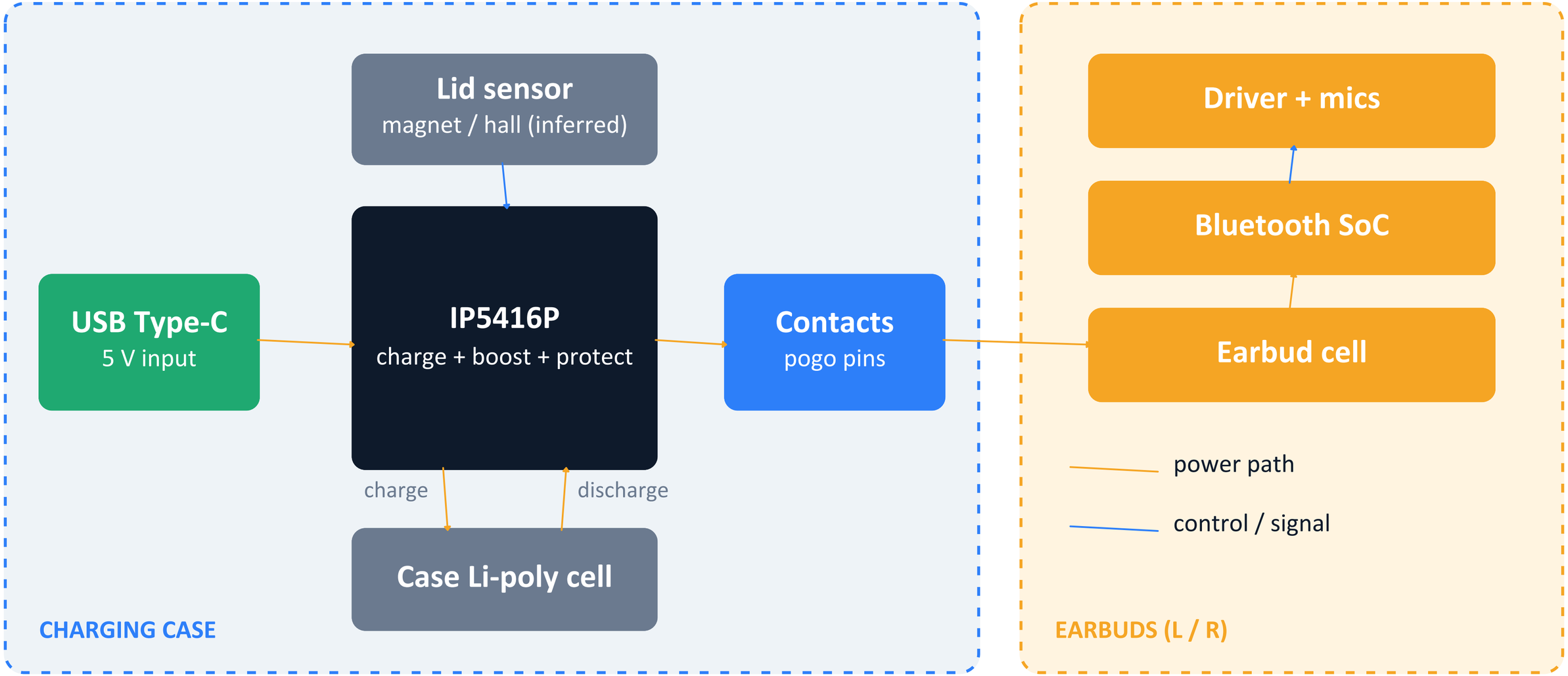
Charge contacts

Metal pads on the stem receive 5 V from the case

Lead wires

Colored wires link the cell, driver and board

Internal block diagram (functional view)



Key subsystem: IP5416P power management

4.1 Function

Charges the case cell and boosts it to 5 V for the earbuds.

4.2 Working principle

Regulates charge into the Li-polymer cell; boost stage steps cell voltage up.

4.3 Characteristics

Highly integrated, few parts, built-in protections.
Ratings: [TBC] from datasheet.

4.4 Role in product

Heart of the case: replaces separate charger, boost and control ICs.

Complete working flow, step by step

1. Plug 5 V into the Type-C port
2. IP5416P charges the case cell
3. Open lid: sensor wakes the IC
4. Insert earbud: load is detected
5. Boost 5 V flows through contacts
6. Earbud cell charges; IC stops at full
7. Close lid: case returns to standby
8. Remove buds: Bluetooth audio plays

Components, function & specifications

Component	Function	Specification
Case PCB + Type-C	5 V input, board carrier	USB-C, 5 V
IP5416P SoC	Charge, boost, protection	Injoinic; ratings [TBC]
Boost inductor	Boost energy storage	Value [TBC]
Case Li-polymer cell	Stores case energy	3.7 V nominal; mAh [TBC]
Lid sensor + magnet	Lid open / close detect	Inferred from photos
Earbud PCB	Bluetooth + charge circuit	SoC model [TBC]
Earbud cell (x2)	Powers each earbud	3.7 V nominal; mAh [TBC]
Dynamic driver (x2)	Electrical to sound	Diameter [TBC]
MEMS microphone	Voice pickup	Type [TBC]
Shells + ear tips	Protection, fit, seal	White plastic, silicone

To complete the report

Items marked [TBC] need datasheet values or measurement.

- Confirm IP5416P ratings from the Injoinic datasheet
- Read cell capacities from labels
- Measure inductor and verify IC marking